

① Define the boundary matrices (aka "incidence matrix")

∂_p: p-simps → (p-1 simplex)

σ_i is incident to σ_j (gen. coef: write the coef of σ_i in $\partial(\sigma_j)$)

$$\partial_k(\sigma) = \sum_{k=0}^{\dim\sigma} (-1)^k [v_0, \ldots, \hat{v}_k, \ldots, v_{\dim\sigma}]$$

($\hat{v}_k$: remove v_k)

② Use matrix-vector multiplication to compute

$\partial_2([v_0 v_1 v_2] + [v_2 v_3 v_0])$

and

$\partial_1([v_0 v_3] + [v_3 v_4] + [v_3 v_2] + [v_2 v_0])$

⓪ $K =$

Today: $\mathbb{Z}_2$-coef.

$K_4 = \emptyset$

$K_3 = \emptyset$

$K_2 = \{ [V_0, V_1, V_2], [V_0, V_2, V_3], [V_1, V_2, V_3], [V_1, V_0, V_3] \}$ (f_1, f_2, f_3, f_4)

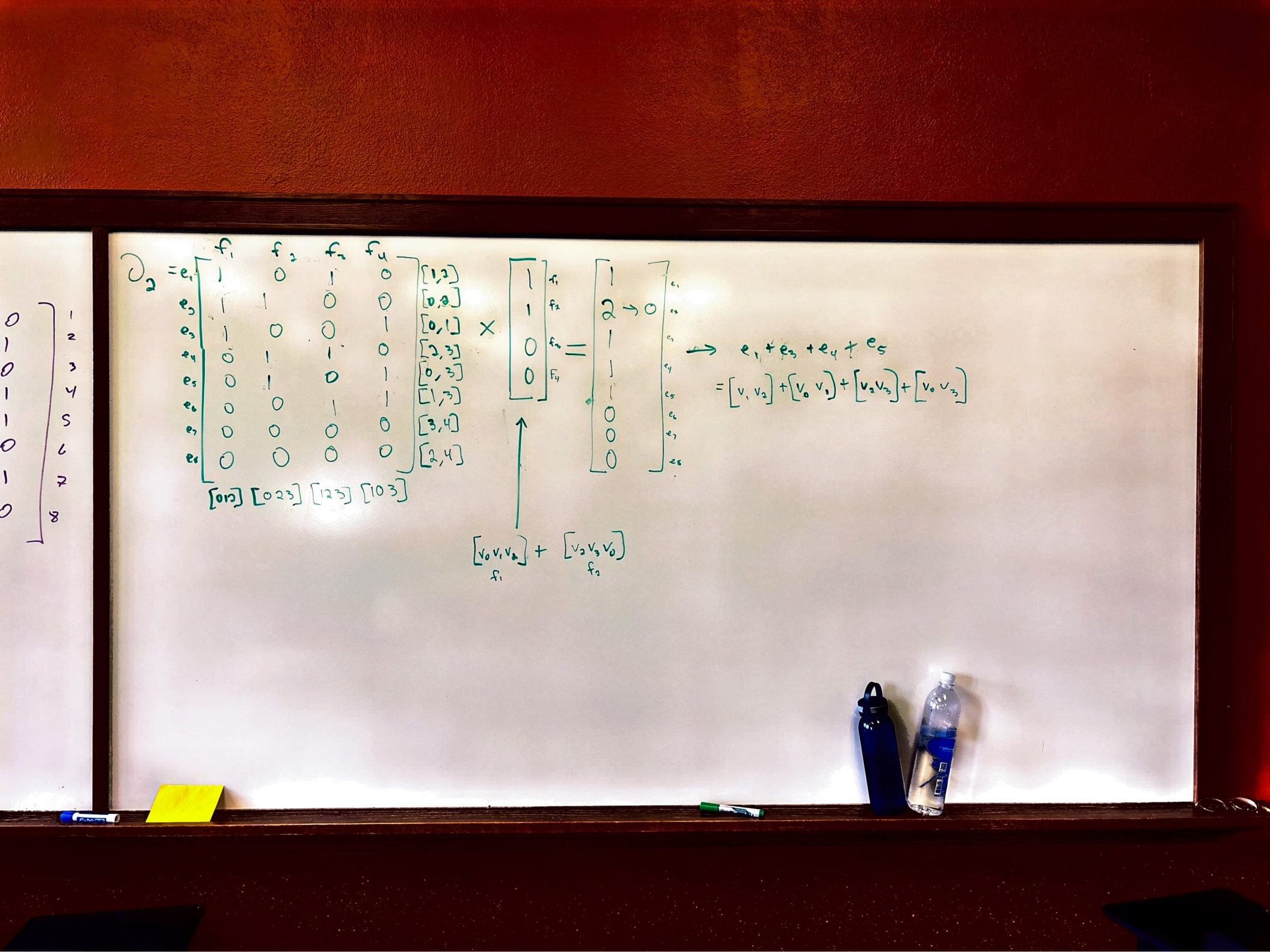
$K_1 = \{ [V_1, V_2], [V_2, V_0], [V_0, V_1], [V_2, V_3], [V_0, V_3], [V_1, V_3], [V_3, V_4], [V_2, V_4] \}$ (e_1, e_2, e_3, e_4, e_5, e_6, e_7, e_8)

$K_0 = \{ v_0, v_1, v_2, v_3, v_4 \}$ (e_2, e_4, e_5, e_7)

$v_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

	e_1	e_2	e_3	e_4	e_5	e_6	e_7	e_8
V_0	0	1	1	0	1	0	0	0
V_1	1	0	1	0	0	1	0	0
V_2	1	1	0	1	0	0	0	1
V_3	0	0	0	1	1	1	1	0
V_4	0	0	0	0	0	0	1	1

$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} = v_3 + v_4$


∂_2
f_1 f_2 f_3 f_4
e_1 1 0 1 0 [1,2]
e_2 1 1 0 0 [0,2]
e_3 1 0 0 1 [0,1]
e_4 0 1 1 0 [2,3]
e_5 0 1 0 1 [0,3]
e_6 0 0 1 1 [1,3]
e_7 0 0 0 0 [3,4]
e_8 0 0 0 0 [2,4]
[013] [023] [123] [103]
×
1 f_1
1 f_2
0 f_3
0 F_4
=
1
2 → 0
1
1
1
0
0
0
e_1 e_2 e_3 e_4 e_5 e_6 e_7 e_8
$\rightarrow e_1 + e_3 + e_4 + e_5$
$= [v_1 v_2] + [v_0 v_1] + [v_2 v_3] + [v_0 v_3]$
$[v_0 v_1 v_2]$ f_1 + $[v_2 v_3 v_0]$ f_2
0 1 0 1 1 0
1 2 3 4 5 6 7 8

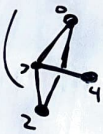
(2) Use matrix-vector multiplication to compute

$$\partial_2\left([v_0 v_1 v_2] + [v_2 v_3 v_0]\right) = [v_1 v_2] + [v_0 v_1] + [v_2 v_3] + [v_0 v_3]$$

and

$$\partial_1\left([v_0 v_3] + [v_3 v_4] + [v_3 v_2] + [v_2 v_0]\right) = v_3 + v_4$$

$$\partial\left(\right) = v_3 \quad v_4$$



Question:

For p prime, $\mathbb{Z}_p$ is a field.

What are the two operations?

(try for small p)

Matrix Reductions / Gaussian Elimination

- add rows
- add columns
- swap rows/columns
- multiply by a non-zero scalar

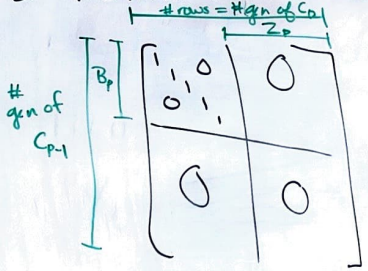
↑ preserves rank + nullspace

Do the following:

(1) Use col. addition to "zero out" as many columns as you can.

(2) Use row addition to "zero out" as many rows as you can

(3) Swap rows/columns to get a matrix that looks like this:



"Smith normal form"

☝ include row + colum labels when you add

a_2 has 1 zero cols.

a_1 has 4 zero cols.